

INFRARED ABSORPTION SPECTROSCOPY

What is spectroscopy?

- Spectroscopy is an **analytical technique** which helps to determine structure.
- It is basically the study of **the interaction of matter and electromagnetic radiation**.
- It destroys **little or no sample**.

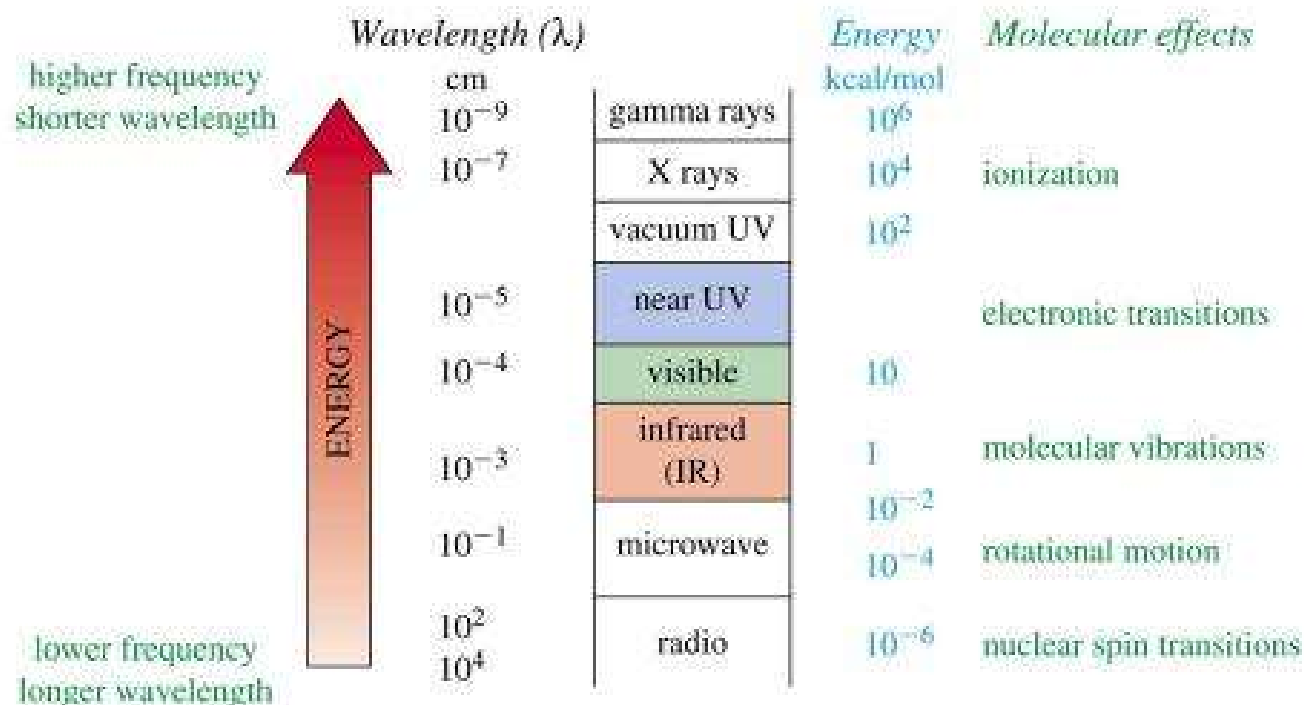
Types of Spectroscopy

- **Infrared (IR) spectroscopy** measures the bond vibration frequencies in a molecule and is used to determine the functional group.
- **Mass spectrometry (MS)** fragments the molecule and measures the masses.
- **Nuclear magnetic resonance (NMR) spectroscopy** detects signals from hydrogen atoms and can be used to distinguish isomers.
- **Ultraviolet (UV) spectroscopy** uses electron transitions to determine bonding patterns.

Early history of the study of infrared absorptions

- The most important pioneer of IR spectroscopy was **William W. Coblentz**. In 1905 he published the result of a large study of compounds whose spectra he had recorded from **1000 nm to 16,000 nm**. Coblentz's work was a breakthrough in that researchers were able to relate the **character of groups of atoms within molecules as being related to specific absorptions** in the mid-IR (2500–25,000 nm).
- These absorptions are the result of interactions with the **fundamental vibrations of the chemical bonds** associated with the atoms of the groups.

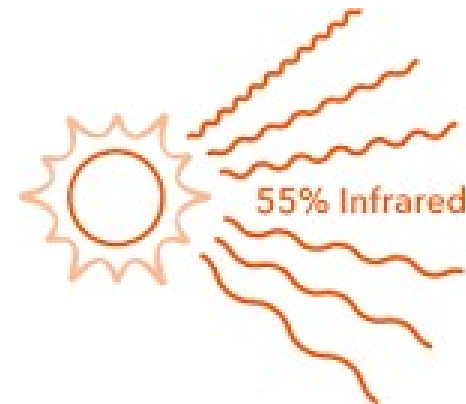
The Spectrum and Molecular Effects



The IR Region

- Just below red in the visible region.
- Wavelengths between 800 nm and 25 μm (usually **2.5-25 μm**).
- More common units are wavenumbers, or cm^{-1} , the reciprocal of the wavelength in centimeters.
- Wavenumbers are proportional to frequency and energy.

The energy of solar radiation at the earth's surface is divided as follows: 6.8% UV (0.5% UVB and 6.3% UVA), 38.9% VIS, and 54.3% NIR light.



Infrared spectroscopy: Principles

We can think of **chemical bonds as weak springs** holding together two or more atoms, these springs will vibrate naturally and when energy is added to the system then they will vibrate more energetically. **However, atoms in molecules are constrained by quantum mechanics so that only a few specific energy levels are allowed.**

If we have only two atoms then **the only vibration will be seen as a stretching**. When three or more atoms are involved then bonds can also **bend**, giving rise to a whole series of different vibrations. **Stretch vibrations require more energy than bending vibrations** but there will also be variation in the energy requirements of the bending vibrations.

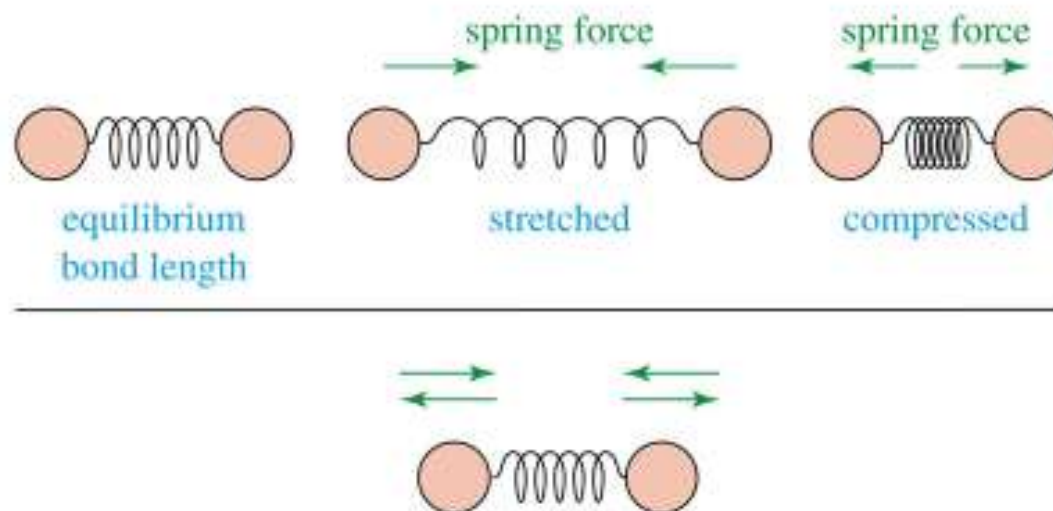
Different chemical bonds (like O–H, C–H and N–H) **vary in strength and hence the amount of energy required for the bond vibration to move from one level to the next**. **This variation in energy will be seen in a spectrum as a series of absorptions at different wavelengths**. By looking at the spectrum we can deduce what vibrations are occurring and hence work out the structure of the molecule (or groups of atoms present).

Principles

- When the certain molecules are exposed to IR light, it will absorb the energy to make the atoms of molecules vibrate.
- The vibration frequency depend on the **mass of atoms, bond length and the bond strength**. (The **stronger** the bond, the **greater** the energy required to stretch it. The frequency of the vibration is **inversely proportional** to the mass of the atoms, so heavier atoms vibrate at lower frequencies.)
- Molecule vibration is stimulated by absorption of radiation in the same frequency with the frequency of its origin vibrations.

Molecular Vibrations

Covalent bonds vibrate at only certain allowable frequencies.



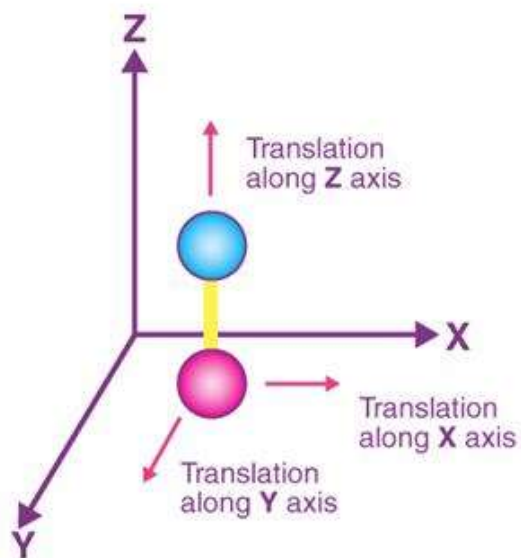
Covalent bonds in organic molecules are not rigid sticks – rather, they behave more like springs. **At room temperature, organic molecules are always in motion**, as their bonds stretch, bend, and twist. The energy of molecular vibration is **quantized rather than continuous**, meaning that a molecule can only stretch and bend at certain 'allowed' frequencies.

Modes of vibrations (normal modes)

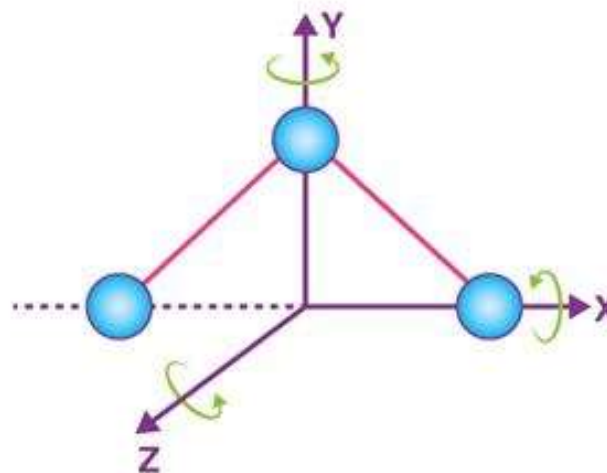
- A molecule has as many degrees of freedom as the total degrees of freedom of its individual atoms. Each atom has **three degrees of freedom corresponding to the Cartesian coordinates (x, y, z) necessary to describe its position relative to other atoms** in the molecule. A molecule of **n atoms** therefore has **3n degrees of freedom**.
- For nonlinear molecules, three degrees of freedom describe **rotation** and three describe **translation**; the remaining **3n - 6 degrees of freedom** are vibrational degrees of freedom or fundamental vibrations.
- Linear molecules have **3n - 5 vibrational degrees of freedom**, for only **two degrees of freedom** are required to describe **rotation**. (For a linear molecule, rotation around its own axis is no rotation because it leave the molecule unchanged)

Modes of vibrations (normal modes)

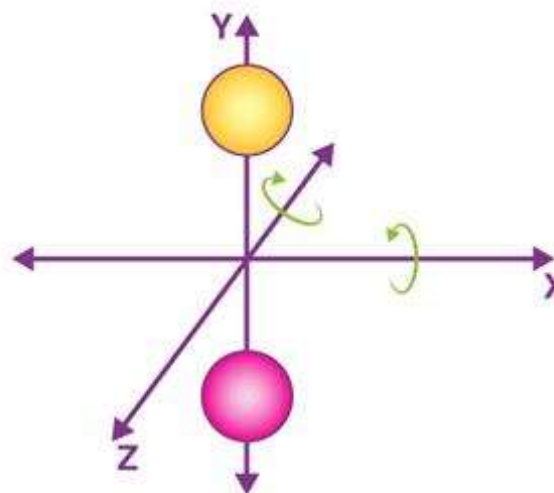
Translational:



Rotational (Non linear molecule):

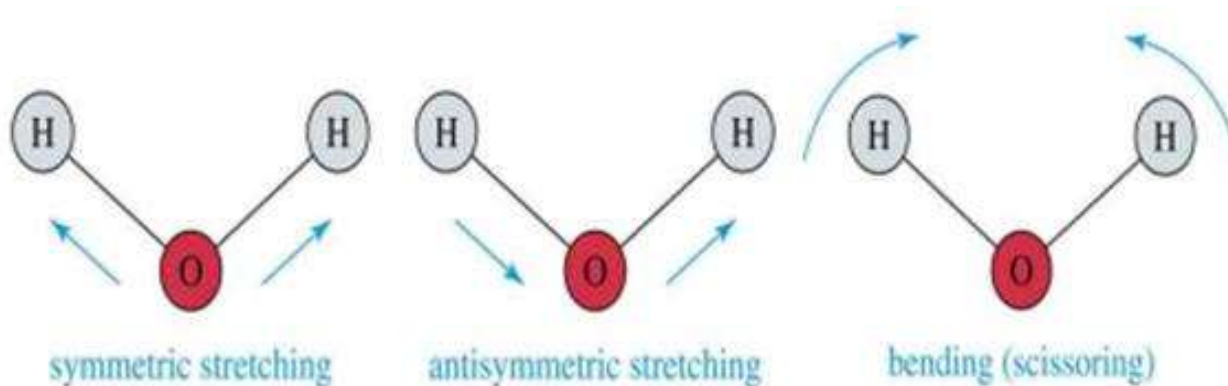


Rotational (linear molecule):



Number of Fundamental Vibration

- A normal 3-atom linear molecule such as carbon dioxide (CO_2) has $(3 \times 3) - 5 = 4$ **normal modes**. The normal modes of CO_2 are a symmetric stretch, an asymmetric stretch, and two bending vibrations.
- Water is a nonlinear molecule, and has $(3 \times 3) - 6 = 3$ **normal modes**. The normal modes of H_2O consists of a symmetric stretch, an asymmetric stretch, and one bending vibration.
- When a molecule absorbs IR light and begins vibrating, it gives rise to an absorption band in the IR spectrum of the molecule at the wavenumber of light absorbed.



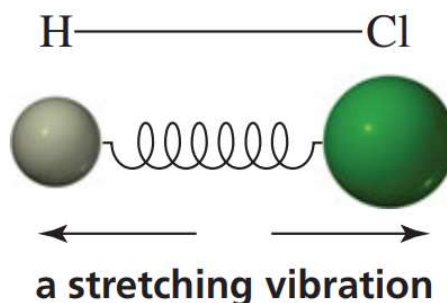
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The length reported for a bond between two atoms is an average length, because in reality a bond behaves as if it were a vibrating spring. A bond vibrates with both stretching and bending motions.

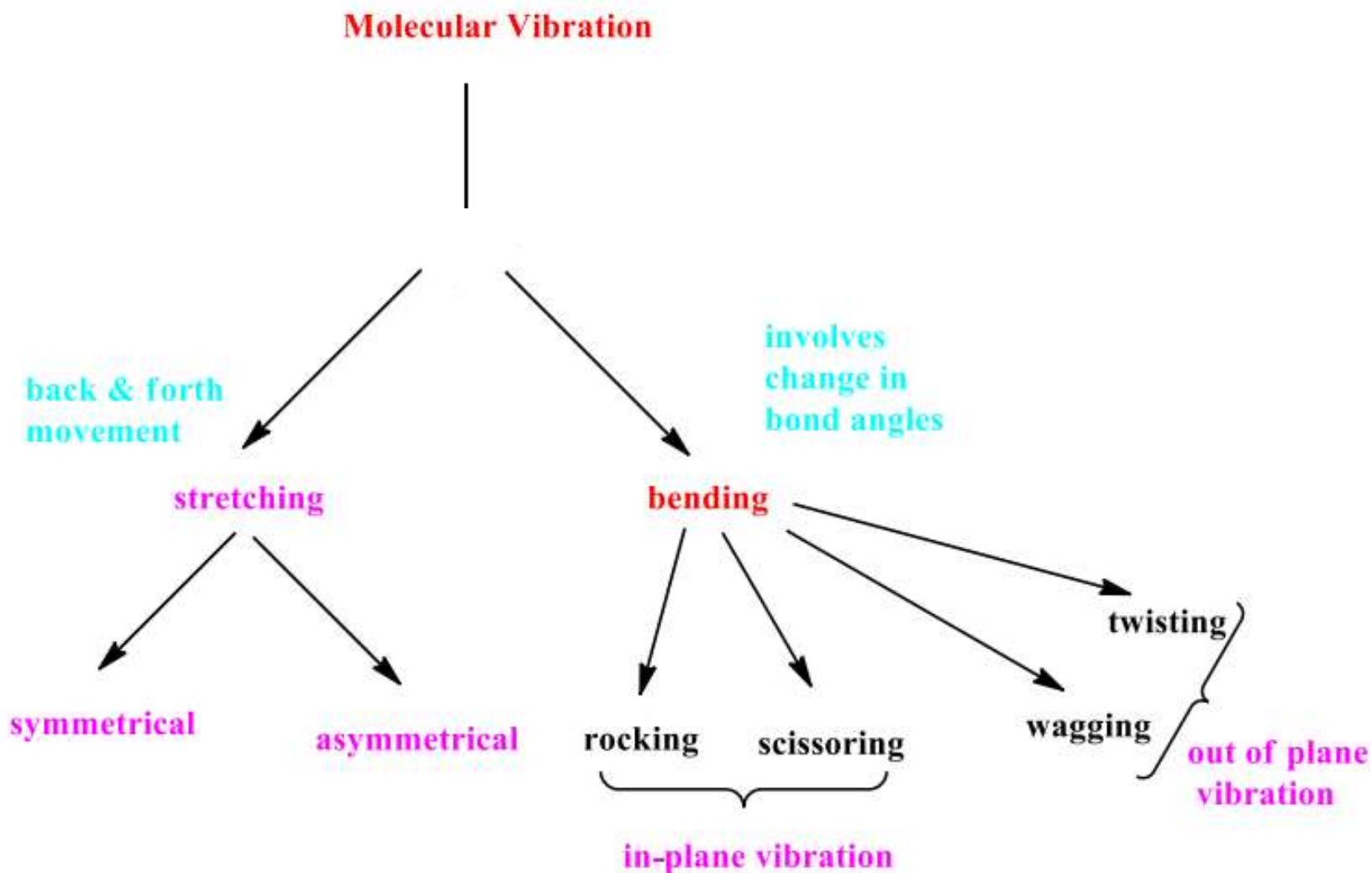
A *stretch* is a vibration occurring along the line of the bond; a stretching vibration changes the bond length.

A *bend* is a vibration that does *not* occur along the line of the bond; a bending vibration changes the bond angle.

A diatomic molecule such as H—Cl can undergo only a **stretching vibration** because it has no bond angles.



Fundamental vibrations involve no change in the centre of gravity of the molecule.



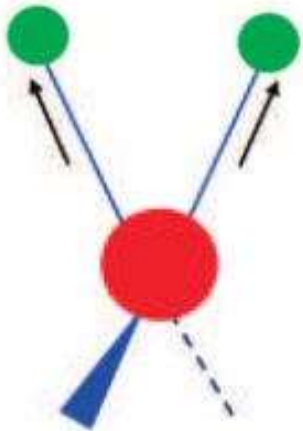
Stretching Vibration

- A stretching vibration is a rhythmical movement along the bond axis such that the inter-atomic distance is increasing or decreasing.
- In this type of vibrations, the distance between the atoms increases or decreases but the atoms remain in the same bond axis.
- **Types of stretching vibrations:** There are two types of stretching vibrations:
 1. **Symmetric stretching:** In this type, the movement of the atoms with respect to a particular atom in a molecule is in the same direction. Therefore, in this mode of vibration both the bonds expand or contract at the same time.
 2. **Asymmetric stretching:** In this vibrations, one atom approaches the central atom while the other departs from it. Therefore, in this mode of vibration, one bond expands while the other contracts.

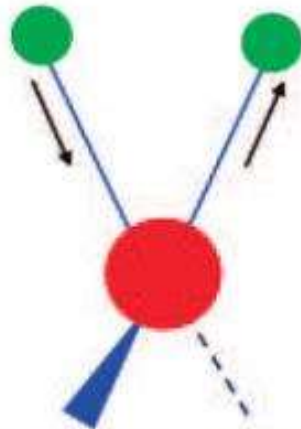
Bending Vibration

- In this type of vibrations, the positions of the atoms change with respect to the original bond axis.
- It consists of a change in bond angle between bonds with a common atom or the movement of a group of atoms with respect to the remainder of the molecule without movement of the atoms in the group with respect to one another.
- **Types of bending vibrations (Deformations):** Bending vibrations are of four types:
 1. **Scissoring:** In this type, two atoms approach each other.
 2. **Rocking:** In this type, the movement of the atoms takes place in the same direction.
 3. **Wagging:** Two atoms move 'up and down' the plane with respect to the central atom.
 4. **Twisting:** In this type, one of the atoms moves up the plane while the other moves down the plane with respect to the central atoms.

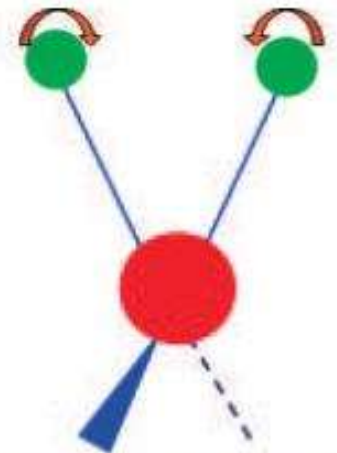
Types of Vibrations



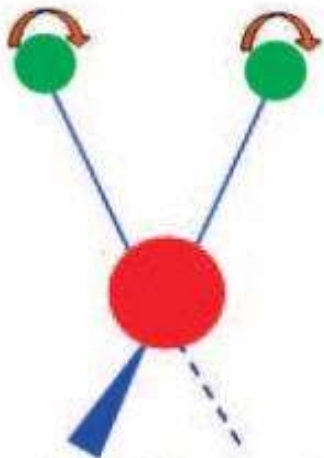
Symmetric



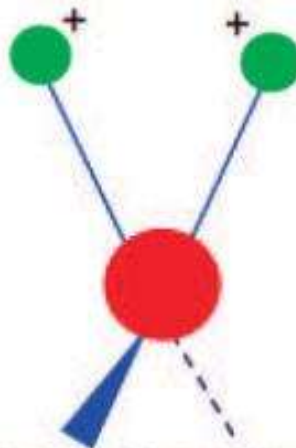
Antisymmetric



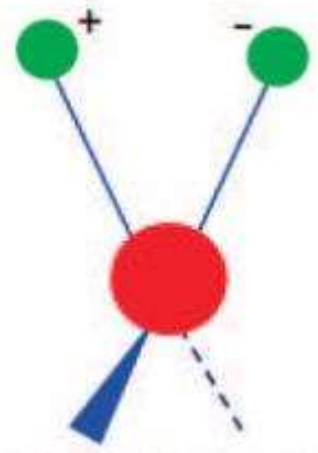
Scissoring (In-plane)



Rocking (In-plane)

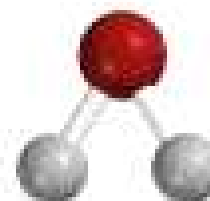
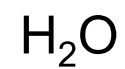
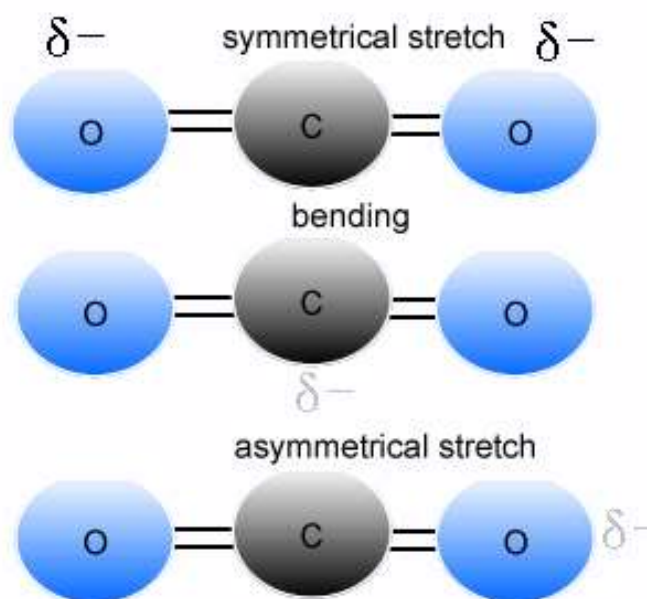


Wagging (Out-of-plane)

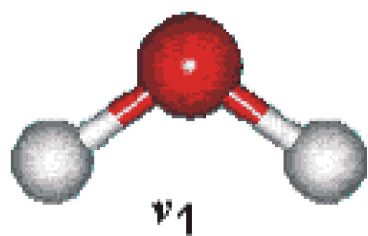
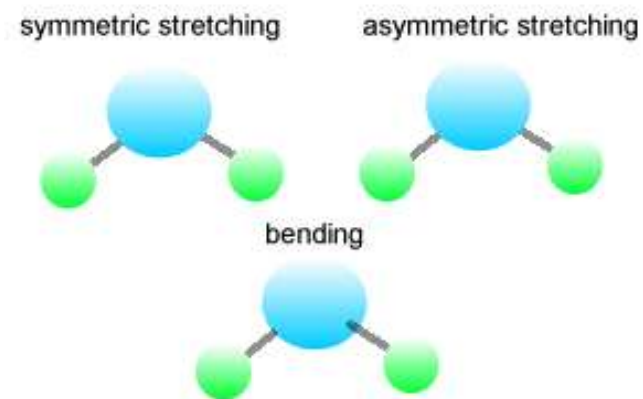
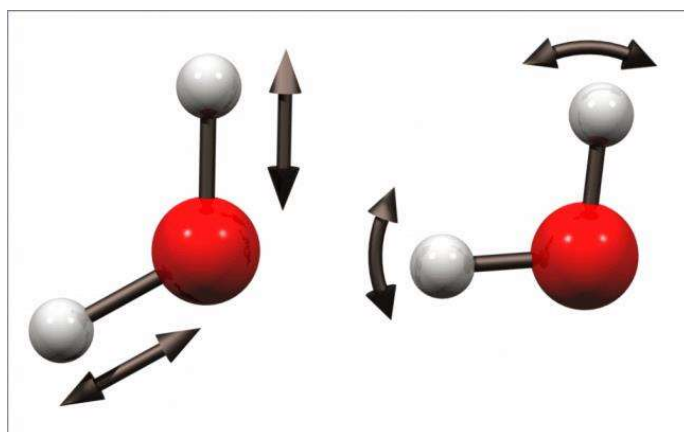


Twisting (Out-of-plane)

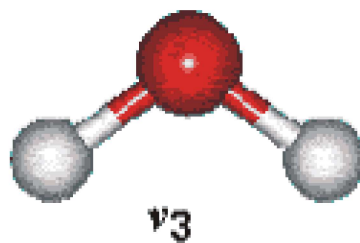
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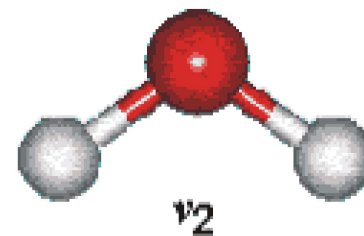
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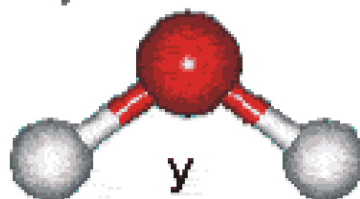
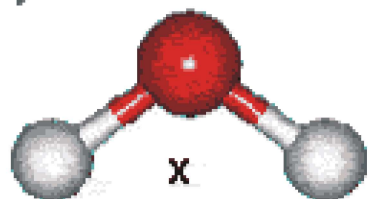
symmetric stretch



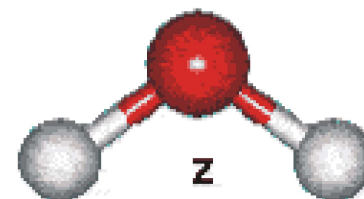
asymmetric stretch



bend

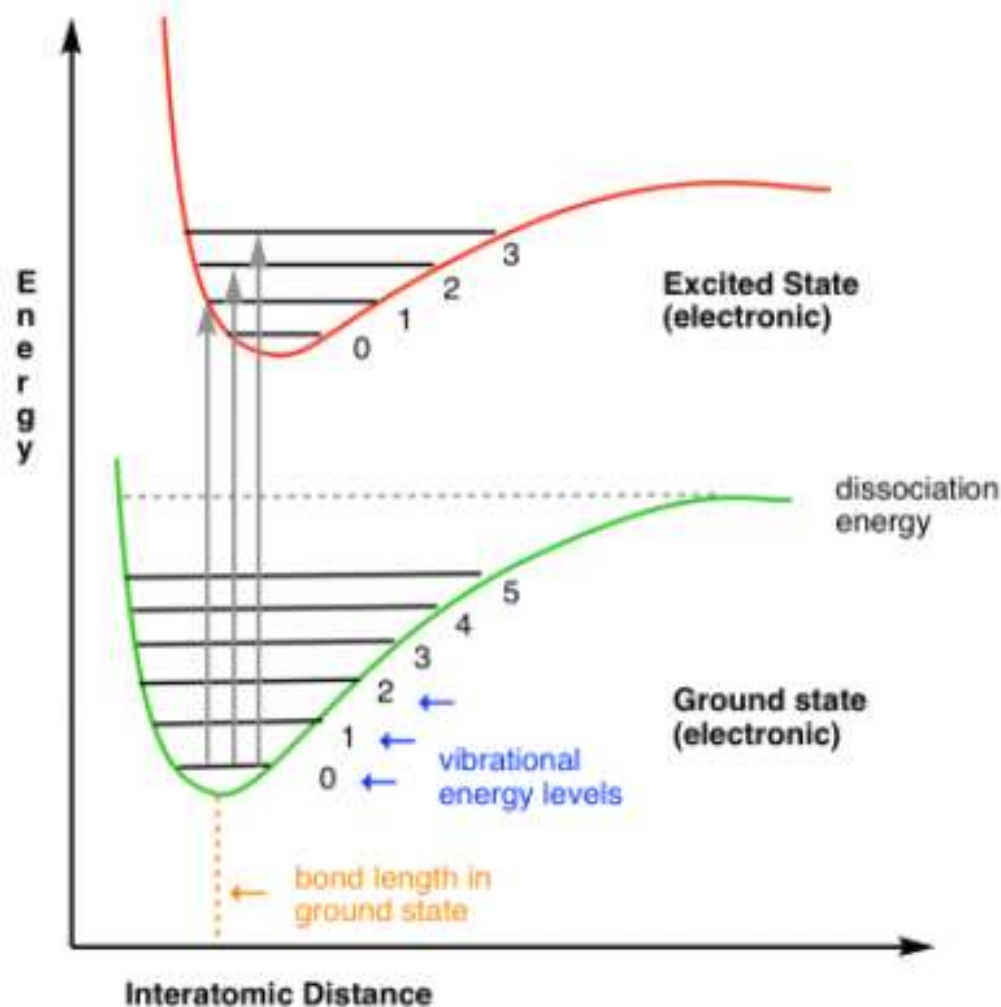


librations



- **When IR absorption occur?**

1. IR absorption occurs when IR radiation interacts with a molecule undergoing a change in dipole moment as it vibrates. As the molecule vibrates, there is a fluctuation in its dipole moment; this causes a field that interacts with the electric field associated with radiation.
2. Infrared absorption only occurs when the incoming IR photon has sufficient energy for the transition to the next allowed vibrational state.



Note: If the 2 rules above are not satisfied, **NO ABSORPTION can occur**

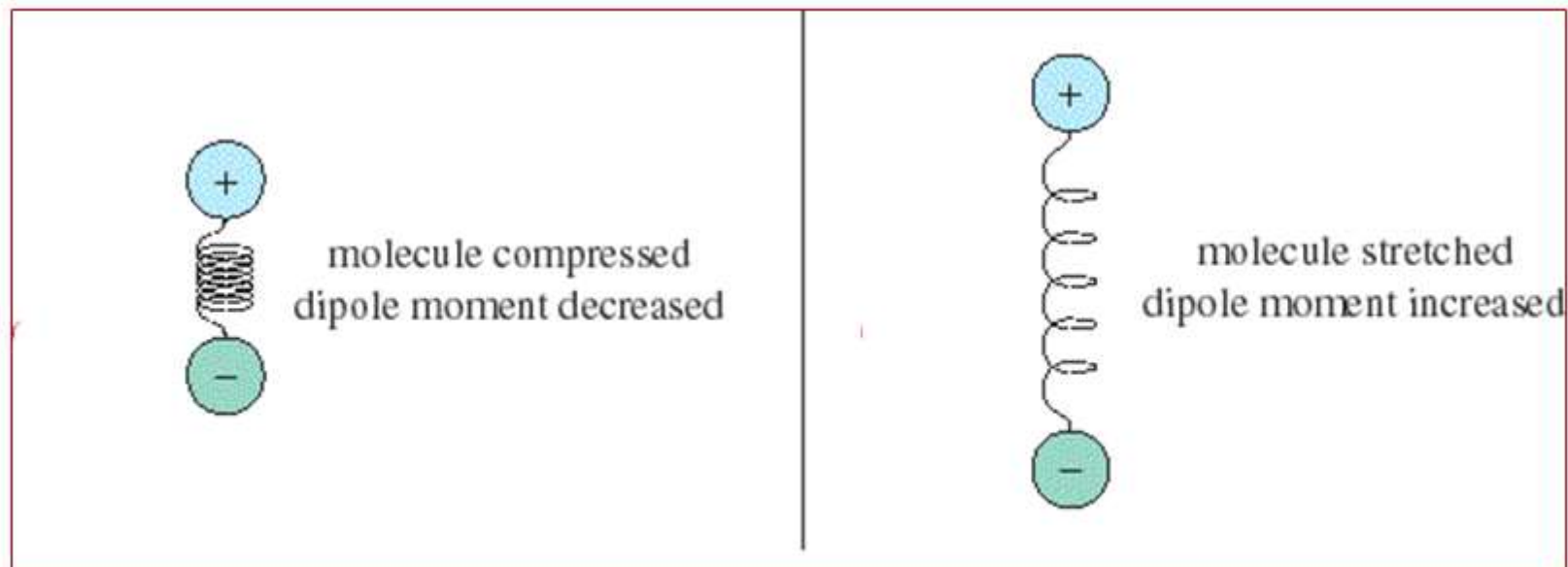
- **NOT ALL** bonds in a molecule are capable of absorbing IR energy. Only those bonds that changes their dipole moment are capable of absorbing IR radiation.
- **The larger the dipole change, the stronger the intensity of the band in an IR spectrum**

IR-Active and Inactive

- A **polar bond** is usually **IR-active**.
- A **nonpolar bond** in a symmetrical molecule **will absorb weakly or not at all**.

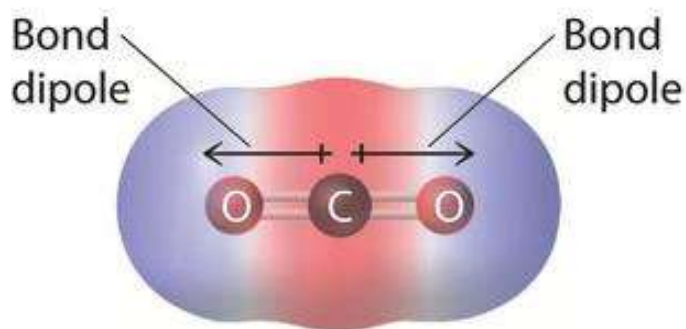
Dipole moment change during vibration:

Dipole moments occur when there is a separation of charge. They can occur between two ions in an ionic bond or between atoms in a covalent bond; dipole moments arise from differences in electronegativity. The larger the difference in electronegativity, the larger the dipole moment. The distance between the charge separation is also a deciding factor in the size of the dipole moment.

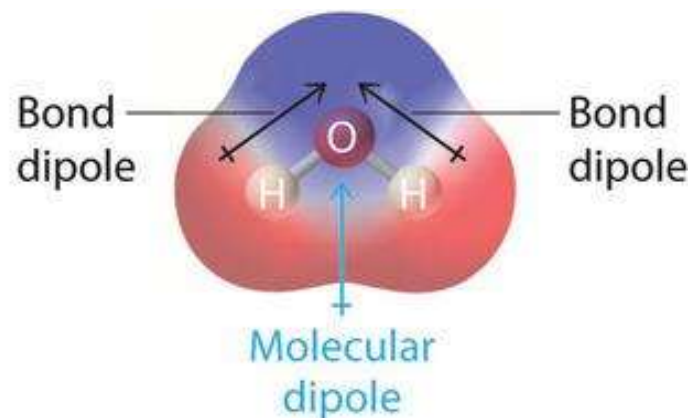


Polarity and Structure of Molecules

- The shape of a molecule and the polarity of its bonds determine the **OVERALL POLARITY** of that molecule. A molecule that contains **polar bonds** might not have any overall polarity, depending upon its shape.
- The simple definition of whether a complex molecule is polar or not depends upon whether its **overall centers of positive and negative charges overlap**. If these centers lie at the same point in space, then the molecule has no overall polarity (and is non polar).
- If a molecule is **completely symmetric**, then the dipole moment vectors on each molecule will cancel each other out, making the molecule nonpolar. A molecule can only be polar if the structure of that molecule is not symmetric.



(a) No net dipole moment

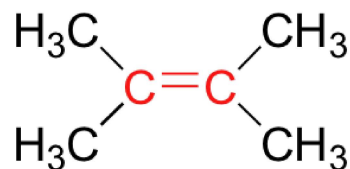


(b) Net dipole moment

IR-Active and Inactive

Some bonds absorb infrared light more strongly than others, and some bonds do not absorb at all. In order for a vibrational mode to absorb infrared light, **it must result in a periodic change in the dipole moment of the molecule**. Such vibrations are said to be **infrared active**. In general, the greater the polarity of the bond, the stronger its IR absorption. **The carbonyl bond is very polar, and absorbs very strongly.** The **carbon-carbon triple bond** in most alkynes, in contrast, is much less polar, and thus a stretching vibration does not result in a large change in the overall dipole moment of the molecule. Alkyne groups absorb rather weakly compared to carbonyls.

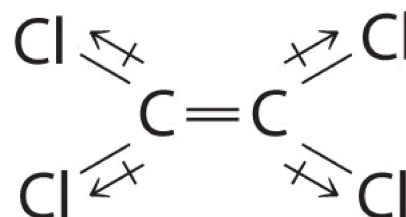
Some kinds of vibrations are **infrared inactive**. **The stretching vibrations of completely symmetrical double and triple bonds, for example, do not result in a change in dipole moment**, and therefore do not result in any absorption of light (but other bonds and vibrational modes in these molecules *do* absorb IR light).



infrared-inactive double and triple bonds

IR-Active and Inactive

If a molecule has a **centre of symmetry**, then the vibrations are centrosymmetric and are inactive in the IR. The vibrations which are not centrosymmetric are active in the IR.

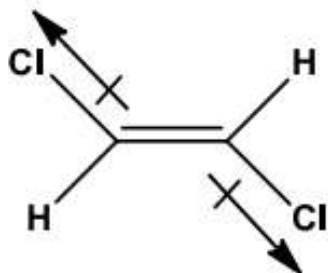


The symmetrical stretching of the C=C bond in ethylene (a molecule with a centre of symmetry) will not produce a change in the dipole moment of the molecule. This mode of vibration is IR inactive. Therefore, ***trans*- dichloroethylene shows no C=C stretching vibration whereas cis-isomer shows this band.**

The symmetrical stretching mode of carbon dioxide produces no change in the dipole moment (μ) of the molecule and is, therefore, 'inactive' in the IR. The **asymmetrical stretching mode produces a change in the dipole moment, is thus IR active.**



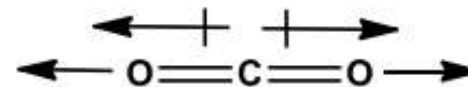
Examples of IR Active/Inactive Molecules/Modes



trans-dichloroethylene



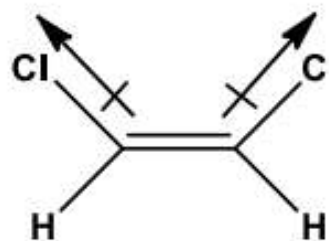
Acetylene



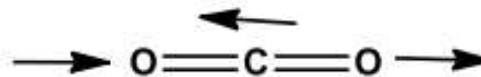
Carbon dioxide

Symmetrical
stretching

IR Inactive ($\mu = 0$)



cis-dichloroethylene



Carbon dioxide

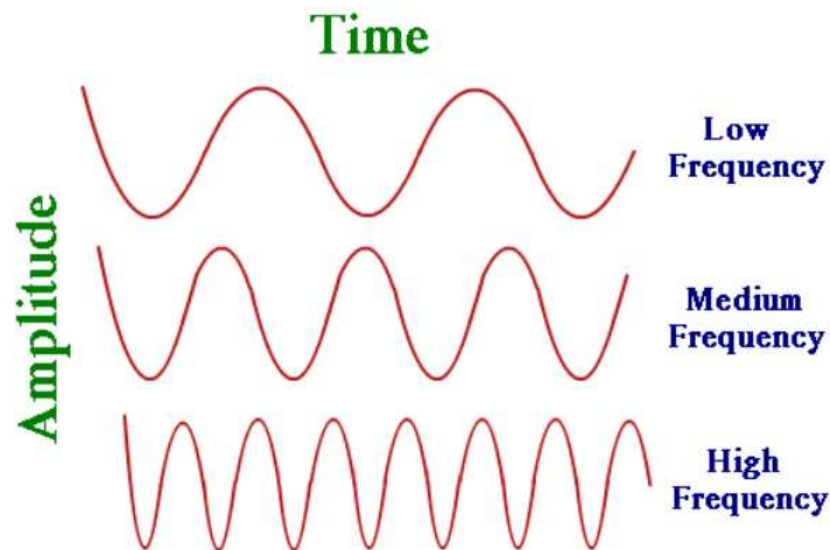
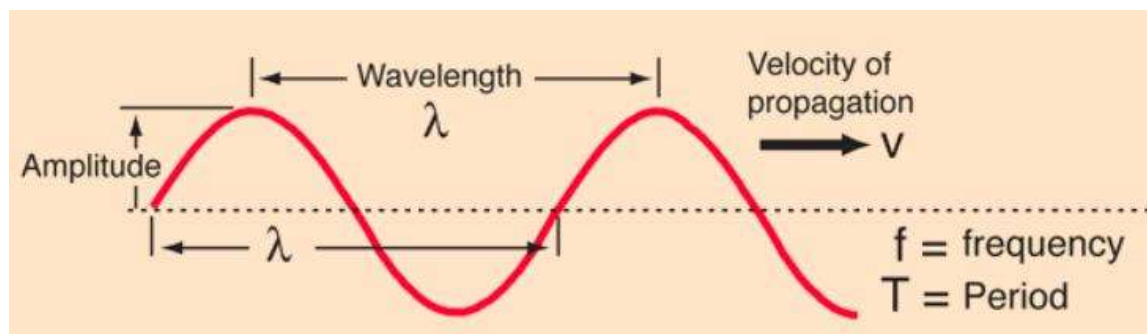
Asymmetrical
stretching

IR Active ($\mu \neq 0$)

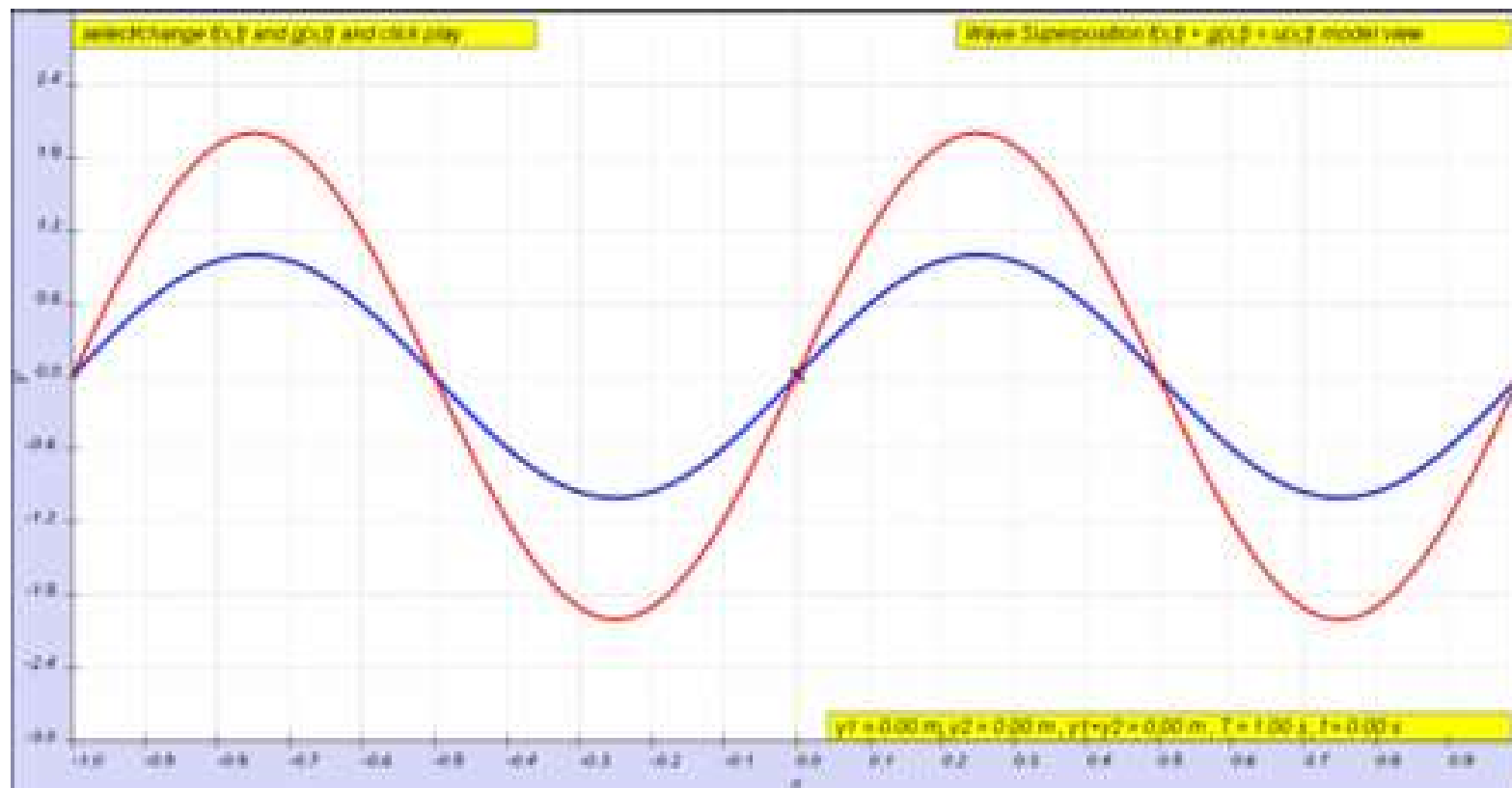
Infrared spectroscopy

The energy of molecular vibration is **quantized** rather than continuous, meaning that a molecule can only stretch and bend at certain 'allowed' frequencies.

If a molecule is exposed to **electromagnetic radiation that matches the frequency of one of its vibrational modes**, it will in most cases **absorb energy from the radiation and jump to a higher vibrational energy state** - what this means is that the **amplitude of the vibration will increase**, but the **vibrational frequency will remain the same**. The difference in energy between the two vibrational states is **equal to the energy** associated with the wavelength of radiation that was absorbed.



The energy absorbed will **increase the amplitude** of the vibrational motions of the bonds in the molecule but not the **frequency**.



Infrared spectroscopy

The IR region of EMR is commonly divided into three sub-regions; **near-infrared (NIR)**, **mid-infrared (mid-IR; MIR)**, and **far-infrared (far-IR; FIR)**. The reasons for such division are both physical and practical. NIR refers to wavenumber region of **12,500–4000** cm^{-1} (800–2500 nm).

NIR region (12,500–4000 cm^{-1}) is populated exclusively with overtone and combination bands with no presence of fundamental ones.

FIR (10-400 cm^{-1}) bands arise mostly from molecule's skeletal deformations and torsional motions, as well as from inter-molecular modes. These fundamental bands are frequently difficult for analysis, as they are weaker and broader than most MIR bands.

Infrared spectroscopy

- MIR region corresponds to wavenumbers $4000\text{--}400\text{ cm}^{-1}$ (wavelengths $2500\text{--}25,000\text{ nm}$) and is mostly associated with the fundamental vibrational transitions. Hence, the absorption coefficient reach high levels in MIR region. Nevertheless, besides strong fundamental bands, weak overtones and combinations are also observed in MIR spectra, increasing its complexity.
- The MIR bands are usually better resolved and separated than strongly overlapping NIR bands. Typically, MIR region is separated into the upper region mostly populated by X-H stretching bands (c. $3200\text{--}2600\text{ cm}^{-1}$; also called C-H stretching region because of the abundance of such functional groups in organic molecules) and lower region (c. $2000\text{--}400\text{ cm}^{-1}$), in which the majority of deformation bands tend to appear.
- The lower region contains highly structure-specific bands that may be considered a molecular fingerprint that allows to identify a given molecular structure from its MIR spectrum. Accordingly, the lower MIR region is being traditionally referred to as the fingerprint region. Note that IR term is sometimes used in the literature in the meaning of MIR, and it is assumed there that IR region corresponds just to $4000\text{--}400\text{ cm}^{-1}$.

Infrared spectroscopy

Infrared radiation has just the right range of frequencies—4000 to 600 cm^{-1} —to correspond to the frequencies of the stretching and bending vibrations of the bonds in organic molecules. It is just below (to the right of) the “red region” of visible light, meaning that infrared light has lower energy than visible light. (*Infra* is Latin for “below.”)

Each stretching and bending vibration of a given bond occurs with a characteristic frequency. When a molecule is bombarded with radiation of a frequency that exactly matches the frequency of the vibration of one of its bonds, the molecule absorbs energy. This allows the bond to stretch and bend a bit more. By experimentally determining the wavenumbers of the energy absorbed by a particular compound, we can ascertain what kinds of bonds it has. For example, the stretching vibration of a C=O bond absorbs energy with wavenumber $\sim 1700 \text{ cm}^{-1}$, whereas the stretching vibration of an O—H bond absorbs energy with wavenumber $\sim 3400 \text{ cm}^{-1}$ (Figure 14.13).

